**UNIVERSITY OF ILLINOIS URBANA-CHAMPAIGN**

STAT 576: Empirical Process Theory and Weak Convergence — Fall 2025
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# Lecture 8

**Content.** Content/keywords of lecture (very briefly)

1. Finishing up percolation
2. Second moment method

## 1 Finishing up percolation

**Recall:**

$$\Theta(p) := \mathbb{P}_p\big(|C_0| = +\infty\big), \qquad p_c(\mathbb{L}^2) := \sup\{\, p : \Theta(p) = 0 \,\}.$$

**Theorem 1.** $p_c(\mathbb{L}^2) \in (0,1)$

**Proof:** Previously we showed that if $p \le \frac{1}{3}$ then $\Theta(p) = 0$. Therefore $p_c(\mathbb{L}^2) \ge \frac{1}{3}$.

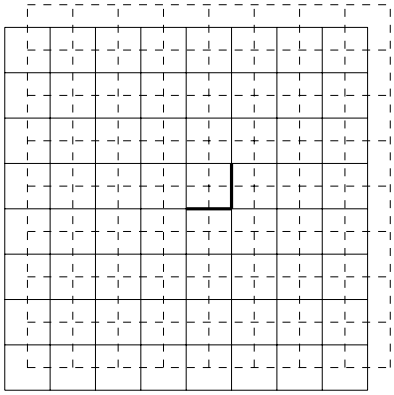
**Dual lattice:** Given $\mathbb{L}^2 = (V, E)$ with $V = \mathbb{Z}^2$, we define the dual lattice

$$\widehat{\mathbb{L}}^2 := \big(\tfrac{1}{2}, \tfrac{1}{2}\big) + \mathbb{L}^2.$$

and $(u, v) \in E$ exactly when $\|u - v\|_1 = 1$. *[Peierls' argument]*

**Fact:** Each edge in the dual lattice $\widehat{\mathbb{L}}^2$ intersects perpendicularly a unique edge in the original lattice $\mathbb{L}^2$.

**Lemma (contour argument):** $|C_0| < \infty$ if and only if there exists a closed cycle surrounding the origin in the dual lattice $\widehat{\mathbb{L}}^2$.



**Claim:** $p_c(\mathbb{L}^2) < 1$
**Proof.**

$$1 - \Theta(p) \;=\; \mathbb{P}_p\big(|C_0| < \infty\big).$$

By the contour lemma,

$$\mathbb{P}_p\big(\exists\, n \geq 4 :\; T_n > 0\big),$$

where $T_n$ denotes the number of closed cycles encircling the origin in the dual lattice $\widehat{\mathbb{L}}^2$.
By the union bound,

$$\mathbb{P}_p\big(\exists\, n \geq 4 :\; T_n > 0\big) \;\leq\; \sum_{n \geq 4} \mathbb{P}_p\big(T_n > 0\big).$$

By Markov's inequality,

$$\sum_{n \geq 4} \mathbb{P}_p\big(T_n > 0\big) \;\leq\; \sum_{n \geq 4} \mathbb{E}[T_n].$$

Bounding the expectation,

$$\mathbb{E}_p[T_n] \;\leq\; \frac{n}{2}\, 3^{\,n-1} (1-p)^n.$$

Hence,

$$1 - \Theta(p) \;\leq\; \sum_{n \geq 4} \frac{n}{2}\, 3^{\,n-1} (1-p)^n.$$

If $p = 1$, the expression above is 0. If $p$ is close to 1, then $1 - \Theta(p) < 1$. This implies $\Theta(p) > 0$. Therefore, $p_c(\mathbb{L}^2) < p^* < 1$.

**Exercise.** Show that for $L^d$ (the $d$-dimensional lattice),

$$p_c(\mathbb{L}^d) \;\geq\; \frac{1}{2d-1}.$$

## 2 Second Moment Method

For an almost surely nonnegative, integer-valued random variable $Z$,

$$\mathbb{P}(Z > 0) \;\le\; \mathbb{E}[Z].$$

**Want:** Lower bound $\mathbb{P}(Z > 0) \ge ...$

**Question:** Does a large $\mathbb{E}Z$ imply that $Z > 0$ with large probability?

**Example:** Let $Z = Z_n$ be

$$Z_n = \begin{cases} n^2, & \text{with probability } 1/n, \\ 0, & \text{with probability } 1 - 1/n. \end{cases}$$

Then $\mathbb{E}[Z_n] = n \to \infty$ while $\mathbb{P}(Z_n > 0) = 1/n \to 0$. [Lottery effect].

**Intuition:** If $Z$ has small variance, then $Z$ is close to $\mathbb{E}Z$.

**Lemma (Chebyshev's Inequality):** For every $\varepsilon > 0$,

$$\mathbb{P}\big(|X - \mathbb{E}X| \ge \varepsilon\big) \;\le\; \frac{\mathrm{Var}(X)}{\varepsilon^2}.$$

**Proof:** By Markov's inequality,

$$\mathbb{P}\big(|X - \mathbb{E}X| \ge \varepsilon\big) = \mathbb{P}\big((X - \mathbb{E}X)^2 \ge \varepsilon^2\big) \le \frac{\mathbb{E}\big[(X - \mathbb{E}X)^2\big]}{\varepsilon^2} = \frac{\mathrm{Var}(X)}{\varepsilon^2}.$$

Then by Chebyshev's inequality,

$$\mathbb{P}(Z = 0) = \mathbb{P}\big(Z - \mathbb{E}Z = -\,\mathbb{E}Z\big) \le \mathbb{P}\big(|Z - \mathbb{E}Z| \ge \mathbb{E}Z\big) \le \frac{\mathrm{Var}(Z)}{(\mathbb{E}Z)^2}.$$

Hence,

$$\boxed{\mathbb{P}(Z > 0) \;\ge\; 1 - \frac{\mathrm{Var}(Z)}{(\mathbb{E}Z)^2}}$$

and in particular, if $\mathrm{Var}(Z) = o\big((\mathbb{E}Z)^2\big)$ then $\mathbb{P}(Z > 0) \ge 1 - o(1)$.

Paley-Zygmund inequality: $Z \ge 0$ a.s. For any $0 \le \theta < 1$,

$$\mathbb{P}\big(Z \ge \theta\,\mathbb{E}Z\big) \;\ge\; (1 - \theta)^2\,\frac{(\mathbb{E}Z)^2}{\mathbb{E}[Z^2]}, \quad \text{so in particular} \quad \mathbb{P}(Z > 0) \;\ge\; \frac{(\mathbb{E}Z)^2}{\mathbb{E}[Z^2]}\,.$$

**Corollary** With $\theta = 0$ in Paley-Zygmund,

$$\mathbb{P}(Z > 0) \;\ge\; \frac{(\mathbb{E}Z)^2}{\mathbb{E}[Z^2]} = \frac{(\mathbb{E}Z)^2}{(\mathbb{E}Z)^2 + \mathrm{Var}(Z)} = 1 - \frac{\mathrm{Var}(Z)}{(\mathbb{E}Z)^2 + \mathrm{Var}(Z)}.$$

*(This corollary gives a stronger lower bound.)*

**Theorem 2.** $m$ balls are thrown independently into $n$ bins.

(a) If $m \geq (1+\varepsilon) n \log n$, then w.h.p. there are no empty bins.

(b) If $m \leq (1-\varepsilon) n \log n$, then w.h.p. there are empty bins.

**Proof.** (a) First-moment method (last time).
(b) Let $Z$ be the number of empty bins. Then

$$Z = \sum_{i=1}^{n} I_i, \qquad I_i = \begin{cases} 1, & \text{if bin } i \text{ is empty,} \\ 0, & \text{otherwise.} \end{cases}$$

Hence

$$\mathbb{E}Z \;=\; n\Big(1 - \frac{1}{n}\Big)^m,$$

and

$$\begin{aligned} Z^2 \;&=\; \sum_{i,j=1}^{n} I_i I_j \\ &=\; \Big(\sum_{i=1}^{n} I_i\Big)\Big(\sum_{j=1}^{n} I_j\Big) \\ &=\; \sum_{i=1}^{n} I_i^2 \;+ \sum_{\substack{i,j=1 \\ i \neq j}}^{n} I_i I_j. \end{aligned}$$

$$\begin{aligned} \mathbb{E}[Z^2] &= \sum_{i=1}^{n} \mathbb{E}[I_i] \;+ \sum_{\substack{i,j=1 \\ i \neq j}}^{n} \mathbb{E}[I_i I_j] \\ &= \mathbb{E}Z \;+\; n(n-1)\,\mathbb{P}(I_i = I_j = 1). \end{aligned}$$

$$\begin{aligned} \mathbb{E}Z &= n\left(1 - \frac{1}{n}\right)^m \\ &= n\,\exp\big(m \ln(1 - \tfrac{1}{n})\big) \\ &= n\,\exp\Big(-\frac{m}{n} + \Theta\Big(\frac{m}{n^2}\Big)\Big). \end{aligned}$$

For $m = (1-\varepsilon) n \log n$,

$$\begin{aligned} \mathbb{E}Z &= n\,\exp\big(-(1-\varepsilon)\log n + \Theta(\tfrac{\log n}{n})\big) \\ &= n^{\varepsilon}\exp\big(\Theta(\tfrac{\log n}{n})\big) \\ &\to\; \infty. \end{aligned}$$

$$\mathbb{P}(I_i = I_j = 1) = \left(1 - \frac{2}{n}\right)^m$$

So

$$\mathbb{E}[Z^2] = \mathbb{E}Z + n(n-1)\left(1 - \frac{2}{n}\right)^m.$$

$$\begin{aligned}\mathbb{P}(Z > 0) \;&\geq\; \frac{(\mathbb{E}Z)^2}{\mathbb{E}[Z^2]} \\ &= \frac{1}{\dfrac{1}{\mathbb{E}Z} + \dfrac{n(n-1)}{n^2} \cdot \dfrac{(1-2/n)^m}{(1-1/n)^{2m}}}\end{aligned}$$

The remianing terms:

$$\begin{aligned}\frac{n(n-1)}{(\mathbb{E}Z)^2}\left(1 - \frac{2}{n}\right)^m &= \frac{n-1}{n}\,\frac{\exp\big(m\ln(1-\frac{2}{n})\big)}{\exp\big(2m\ln(1-\frac{1}{n})\big)} \\ &= \left(1 - \frac{1}{n}\right)\exp\Big(m\Big[\ln\Big(1-\tfrac{2}{n}\Big) - 2\ln\Big(1-\tfrac{1}{n}\Big)\Big]\Big) \\ &= \left(1 - \frac{1}{n}\right)\exp\left(O\left(\frac{n\log n}{n^2}\right)\right)\end{aligned}$$

since $\ln(1-x) = -x + \Theta(x^2)$ as $x \to 0$.

Note that $e^x \sim 1 + x$ as $x \to 0$. So

$$\begin{aligned}\mathbb{P}(Z > 0) &\geq \frac{1}{n^{-\Omega(1)} + (1-1/n)(1+\log n/n)} \\ &= 1 - \theta(\log n/n)\end{aligned}$$

Thus, as $n \to \infty$, $\mathbb{P}(Z > 0) \to 1$. So $\mathbb{E}[Z^2] = (\mathbb{E}[Z])^2(1 + o(1))$.

**Threshold in graphs:** Consider $G \sim G(n,p)$ where the edge probability is $p$;
i.e., each edge is present independently with probability $p$.

$$\mathbb{P}\big(G \text{ has property } \mathcal{P}\big) = \begin{cases}1, & \text{if } p \gg p_n, \\ 0, & \text{if } p \ll p_n,\end{cases} \qquad \text{(phase transition at the critical threshold } p_n),$$

where $p \gg p_n$ means $p/p_n \to \infty$ and $p \ll p_n$ means $p/p_n \to 0$.

**Theorem.** Let $(c_n)_{n\geq 1}$ be any sequence of positive reals with $\lim_{n\to\infty} c_n = \infty$. Then

$$\mathbb{P}\big(G(n,p) \text{ contains a triangle}\big) \longrightarrow \begin{cases} 1, & \text{if } p \geq \dfrac{c_n}{n}, \\ 0, & \text{if } p \leq \dfrac{1}{c_n\, n}. \end{cases}$$

**Heuristic:** number of triangles in $G(n,p)$

Let $Z$ be the number of triangles in the graph.

$$I_T = \begin{cases} 1, & \text{if the three edges among the vertices in } T \text{ are present (a triangle),} \\ 0, & \text{otherwise.} \end{cases}$$

Then

$$Z = \sum_{T:\,|T|=3} I_T, \qquad \mathbb{P}\big((v_1,v_2,v_3) \text{ is a triangle}\big) = p^3.$$

$$\begin{aligned} \mathbb{E}Z &= \sum_{T:\,|T|=3} \mathbb{E}[I_T] = \binom{n}{3} p^3 \\ &= \frac{n(n-1)(n-2)}{6}\, p^3 = \Theta\big(n^3 p^3\big). \end{aligned}$$

If $p \leq \dfrac{1}{c_n n}$ with $c_n \to \infty$, then

$$\begin{aligned} \Theta(n^3p^3) &= O\left(n^3 \cdot \frac{1}{c_n^3 n^3}\right) \\ &= O\left(\frac{1}{c_n^3}\right) \end{aligned}$$

Thus, if $p = o\left(\frac{1}{n}\right)$, then $\mathbb{E}Z = o(1)$. By Markov's inequality,

$$\mathbb{P}[Z = 0] \to 1 \text{ as } n \to \infty$$

Note that $\mathrm{Var}(Z) = \mathrm{Var}(\sum_T I_T)$. Also note that $\sum_{T,T'} \mathrm{Cov}(I_T, I_{T'}) = \mathbb{E}[I_T I_{T'}] - \mathbb{E}[I_T]\mathbb{E}[I_{T'}]$.

So if $|T \cap T'| = \emptyset$, then

$$\mathbb{E}[I_T I_{T'}] = p^6$$

and

$$\mathbb{E}[I_T] = \mathbb{E}[I_{T'}] = p^3$$

If $|T \cap T'| = 1$, then

$$3\binom{n}{3}\binom{n-3}{2}$$

and

$$\mathbb{E}I_T I_{T'} = p^6 = \mathbb{E}I_T \mathbb{E}I_{T'}$$